Università degli studi di Padova

Stochastic Methods for Finance

Report 1: Pricing Call Options with static binomial model

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1 Company description

Tesla, Inc. is an American multinational company operating in the consumer cyclical sector, within the auto manufacturers industry. The company was incorporated in 2003 under the name Tesla Motors, Inc. and changed its name to Tesla, Inc. in February 2017. Tesla is headquartered in Austin, Texas and employs approximately 134,785 full-time employees worldwide. The company focuses on the design, development, manufacturing, leasing, and sale of electric vehicles, as well as energy generation and storage systems. Tesla's activities are organized into two main segments: Automotive and Energy Generation and Storage.

The automotive segment includes the production and sale of fully electric vehicles such as the Model S, Model 3, Model X, and Model Y. In addition to vehicle manufacturing, Tesla provides related services including maintenance, insurance products, and access to its global network of fast-charging stations known as the Supercharger network. The energy generation and storage segment focuses on renewable energy solutions, including lithium-ion battery storage systems such as Powerwall and Megapack, as well as solar energy products such as solar panels and solar roofs. These products are offered to residential, commercial, and industrial customers.

Tesla is led by its co-founder and Chief Executive Officer Elon Musk, who also serves as Director. Other key members of the management team include Vaibhav Taneja, Chief Financial Officer, and Xiaotong Zhu, Senior Vice President of Asia-Pacific and Global Vehicle Manufacturing. The company invests heavily in advanced technologies such as artificial intelligence, autonomous driving systems, battery innovation, and vehicle software development, which are central to its long-term growth strategy.

Tesla's stock is publicly traded on the NASDAQ exchange under the ticker TSLA. As of March 2026, the company has a market capitalization of approximately $1.53T, with annual revenue of about $94.8B and net income of approximately $3.79B over the last twelve months. Several valuation metrics highlight strong investor expectations regarding Tesla's future growth. The company has a trailing price-to-earnings ratio of about 378, a price-to-sales ratio of around 15.2, and a price-to-book ratio of about 18.6. Tesla's beta coefficient of approximately 1.93 indicates that its stock tends to be more volatile than the overall market. Tesla's stock is highly liquid and actively traded in financial markets, with an average daily trading volume of approximately 65M shares.

The company's shareholder base consists of institutional investors, insiders, and retail investors. Among the largest institutional shareholders are major asset management firms such as The Vanguard Group, BlackRock, and State Street Corporation. Institutional ownership contributes to the high liquidity of Tesla's stock and reflects strong interest from global investment funds.

Finally, Tesla currently does not pay dividends, as the company reinvests its profits into research and development, production expansion, and technological innovation.

Due to its large market capitalization, high trading volume, and significant price volatility, Tesla represents an interesting underlying asset for the study of financial derivatives and the application of option pricing models.

2 Pricing a 3-Month Option

In this section we estimate the price of a European call option on Tesla, Inc. (TSLA) with maturity of three months using a one-step binomial pricing model. The procedure consists of estimating the volatility of the underlying asset from historical data, determining the parameters of the binomial tree, computing the risk-neutral probability, and finally evaluating the option price.

2.1 Historical Data and Volatility Estimation

To estimate the volatility of the underlying asset, historical daily data for Tesla stock were downloaded from Yahoo Finance. In particular, daily historical data for the last three months (from 10 December 2025 to 10 March 2026) were collected. An excerpt of the dataset is reported in Table 1.

For each trading day we consider the adjusted closing price S_t, where t represents the trading day and $t \in [0, 61]$. In this case, the adjusted close coincides with the closing price since the company does not currently pay dividends.

Date	**Open**	**High**	**Low**	**Close**	**Adj Close**	**Volume**
3-9-2026	390.05	401.59	381.40	398.68	398.68	66821300
3-6-2026	398.09	402.35	394.21	396.73	396.73	64054600
3-5-2026	401.57	408.62	399.42	405.55	405.55	51925900
3-4-2026	397.85	408.33	394.58	405.94	405.94	68305500
3-3-2026	395.09	396.34	385.39	392.43	392.43	62617300
3-2-2026	390.60	404.54	388.25	403.32	403.32	55088300

Table 1: Excerpt of Tesla historical daily data downloaded from Yahoo Finance

From the time series $\{S_t\}_{t=0}^{61}$ we compute the daily returns

$$r_t = \frac{S_t - S_{t-1}}{S_{t-1}}, \qquad t = 1, \ldots, 61$$

Using this series of returns, we estimate the daily volatility as the sample standard deviation of the returns:

$$\sigma_d = 0.023$$

Since financial models typically use annualized volatility, we rescale the daily volatility assuming 252 trading days in a financial year:

$$\sigma_y = \sigma_d \sqrt{252} = 0.362$$

2.2 Binomial Model Parameters

To price a European option with maturity of three months, we assume that the underlying asset follows a one-step binomial process. In this model, the stock price can either increase

by a factor u or decrease by a factor d during the time interval. The up and down factors are defined as:

$$u = e^{\sigma_y \sqrt{T}} = 1.198 \qquad d = e^{-\sigma_y \sqrt{T}} = 0.834$$

where $T = \frac{3}{12}$ represents the time to maturity (three months) expressed in years.

2.3 Risk Neutral probability measure

Under the assumption of absence of arbitrage, for a binomial model, the risk-neutral probability for an upward movement is q while the probability of a downward movement is $(1-q)$; they are given by:

$$q = \frac{(1+RT)-d}{u-d} \qquad 1-q = \frac{u-(1+RT)}{u-d}$$

Where R denotes the simple compounded risk-free interest rate.

The risk-free rate is approximated using the current CME Term SOFR 3 months retrieved on 10 March 2026:

$$R = 3.67\%$$

Thus, the capitalization and discounting factor using simple compounding are respectively $1+RT$ and $\frac{1}{1+RT}$.

Using the parameters previously computed, we obtain

$$q = 0.48 \qquad 1-q = 0.52$$

2.4 Option pricing

Let us denote the current value of the stock with s, corresponding to the most recent observation in the historical data sample $s = 398.68$, and the strike price of the option with K; we choose the strike AtTheMoney $K = 400$. At maturity T, the payoff of a European call option is given by

$$H_T = \max(S_T - K, 0) = (S_T - K)^+$$

In the one-step binomial model the stock price at maturity can take two possible values:

$$S_T = \begin{cases} s \cdot u & \text{if upward movement occurs} \\ s \cdot d & \text{if downward movement occurs} \end{cases}$$

The corresponding payoffs are summarized below:

	S_T	H_T
Up	477.79	77.79
Down	332.67	0

Under the risk-neutral probability measure, the fair price of the option at time 0 is the discounted expectation of the future payoff under the risk neutral probability measure Q

$$p_0^{\text{call}} = \frac{1}{(1+RT)} E^Q[H_T] = \frac{1}{(1+RT)} [q \cdot (su-K)^+ + (1-q) \cdot (sd-K)^+] = 37.01$$

2.5 Comparison with Market Quotation

To evaluate the accuracy of the model, we compare the theoretical option price with the market quotation of a Tesla call option with strike $K = 400$ and maturity June 18, 2026. The data retrieved from Yahoo Finance on 13 March 2026 are the following:

Contract	Last Trade	Strike	Last Price	Bid	Ask
TSLA260618C00400000	13-Mar-2026 14:19	400	37.19	37.10	37.30

Using the bid-ask spread, we compute the Mid price as

$$p_{0,\text{market}} = \frac{\text{Bid} + \text{Ask}}{2} = \frac{37.10 + 37.30}{2} = 37.20$$

The theoretical price obtained from the binomial model is

$$p_0^{\text{call}} = 37.01$$

The difference between the model price and the observed market mid price is therefore 0.19 USD, corresponding to approximately 0.5%. This small discrepancy can be attributed to model simplifications, the use of historical volatility instead of implied volatility, and market frictions. Overall, the model provides a reasonable approximation of the market value of the option.

3 Pricing a 6-Month Option

Following the same procedure as Section 2, we estimate the price of a European call option on Tesla, Inc. (TSLA) with maturity of six months using a one-step binomial pricing model.

3.1 Historical Data and Volatility Estimation

Daily historical data for the last six months (from 12 September 2025 to 12 March 2026) were collected. From adjusted close price time series $\{S_t\}_{t=0}^{125}$ we compute daily returns r_t. Next, we compute daily and annual (252 trading days) volatility

$$\sigma_d = 0.027 \qquad \sigma_y = 0.426$$

3.2 Binomial Model Parameters

Given $T = \frac{6}{12}$, the time to maturity expressed in years, and the volatility, we are able to compute the up and down factors of the binomial model.

$$u = 1.35 \qquad d = 0.74$$

3.3 Risk Neutral probability measure

Using the current CME Term SOFR 6 months retrieved on 12 March 2026: $R = 3.64\%$ and simple compounding, we can compute the risk neutral probability weights

$$q = 0.45 \qquad 1 - q = 0.55$$

3.4 Option pricing

Current value of the stock	$s = 395$
Strike price ATM	$K = 395$
Stock price at maturity	$\mathrm{S}_T = \begin{cases} 533.95 & \text{if up movement} \\ 292.22 & \text{if down movement} \end{cases}$
Payoff at maturity	$\mathrm{H}_T = \begin{cases} 138.95 & \text{if up movement} \\ 0 & \text{if down movement} \end{cases}$
Price European call option	$p_0^{call} = 62.09$

3.5 Comparison with Market Quotation

The market quotation of a Tesla call option with strike $K = 395$ and maturity September 18, 2026 [data retrieved from Yahoo Finance on 13 March 2026] is the following:

Contract	**Last Trade**	**Strike**	**Last Price**	**Bid**	**Ask**
TSLA260918C00395000	13-Mar-2026 15:18	395	57.60	56.40	56.80

The Mid price is $p_{0,\text{market}} = 56.60$

The model price is approximately 9.7% higher than the market mid price, which can be attributed to the use of historical volatility rather than implied volatility, the simplification of a one-step binomial tree, and market frictions.

4 Conclusions

We observe that the one-step binomial model overestimates the price of the 6-month call option ($\approx 9.7\%$ above the market mid price). This occurs because the stock is assumed to move directly to a single up or down value, exaggerating extreme outcomes and inflating the expected payoff. In contrast, the 3-month option price was much closer to the market ($\approx 0.5\%$ difference), since shorter maturities reduce the effect of this approximation. Multi-step binomial models would improve accuracy for longer maturities by smoothing intermediate paths.